Graph Sketching: Polar Curves

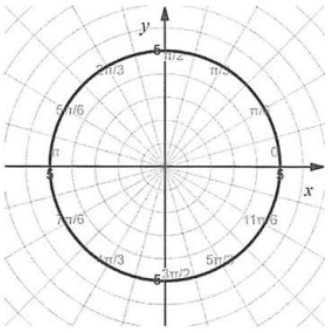
Questions 1–36

Question 1. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$ which are given in polar form:

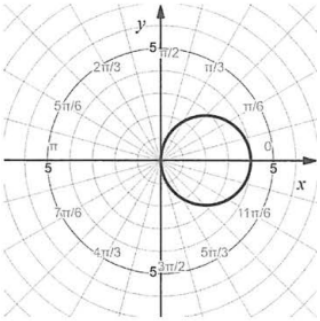
a) $r = 3$
b) $r = \cos\theta$
c) $r = 3\sin\theta$
d) $r = -5\cos\theta$
e) $r = -2\sin\theta$

Question 2. Find the polar equation of each of the following:

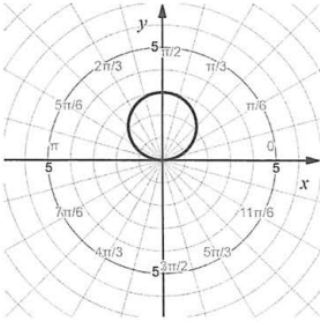
a)



b)



c)



Question 3. Sketch each of the following curves for $r > 0$ which are given in polar form:

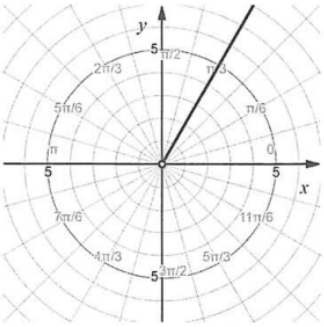
a) $\theta = \frac{\pi}{6}$

b) $\theta = \frac{3\pi}{4}$

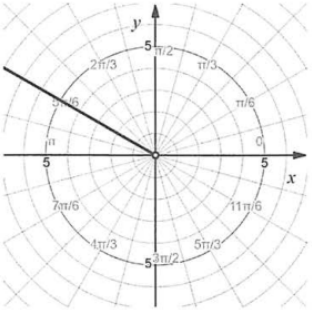
c) $\theta = -\frac{2\pi}{3}$

Question 4. Find the polar equation of each of the following where $r > 0$ and $-\pi < \theta \leq \pi$:

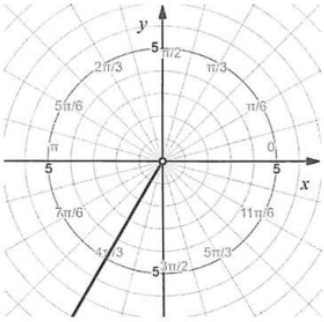
a)



b)



c)



Question 5. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = 3(1 + \cos\theta)$

b) $r = 2(1 + \sin\theta)$

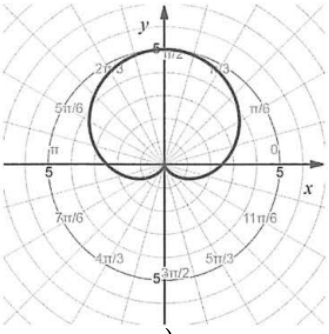
c) $r = 1 - \cos\theta$

d) $r = 4(1 - \sin\theta)$

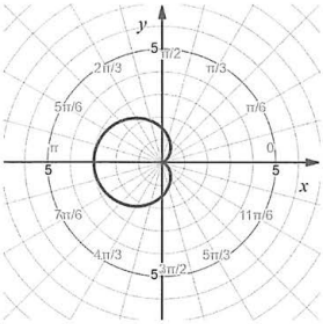
e) $r = 2 + 2\cos\theta$

Question 6. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

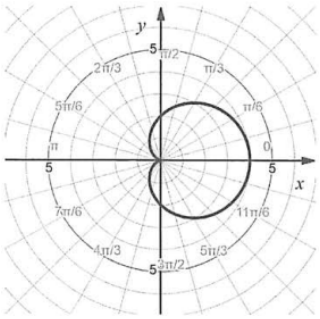
a)



b)



c)



Question 7. Sketch each of the following for $0 \leq \theta \leq 2\pi$, showing any $r < 0$ with a dashed line:

a) $r = 1 + 2\cos\theta$

b) $r = 2 + 4\sin\theta$

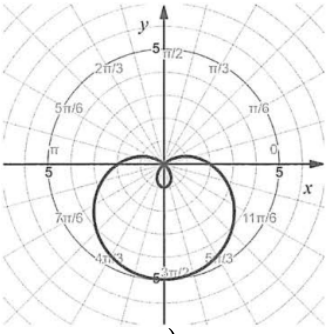
c) $r = 1 + 3\cos\theta$

d) $r = 2 - 3\cos\theta$

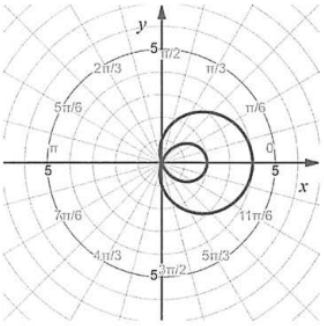
e) $r = 1 - 5\sin\theta$

Question 8. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

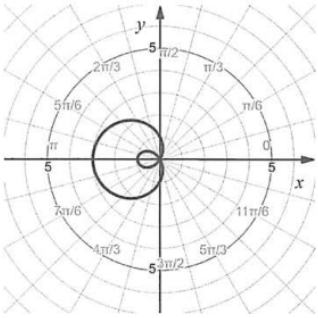
a)



b)



c)



Question 9. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = 2 + \cos\theta$

b) $r = 3 + 2\cos\theta$

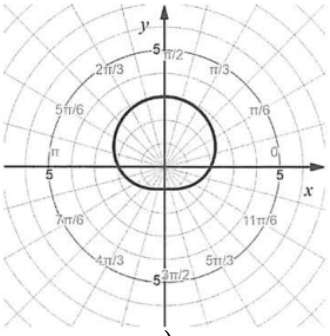
c) $r = 4 + 2\sin\theta$

d) $r = 3 - 2\cos\theta$

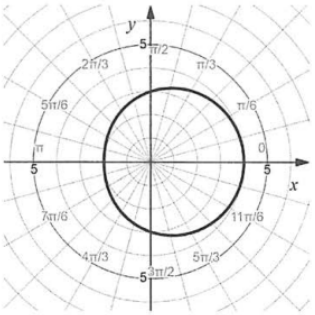
e) $r = \frac{3}{2} - \sin\theta$

Question 10. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

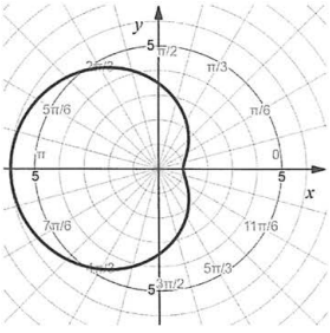
a)



b)



c)



Question 11. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, using a dashed line for $r < 0$:

a) $r = \sin 2\theta$
b) $r = \sin 3\theta$
c) $r = \sin 4\theta$
d) $r = \sin 5\theta$
e) $r = 3\sin 2\theta$
f) $r = 4\sin 3\theta$
g) $r = -\sin 4\theta$
h) $r = -3\sin 5\theta$

Question 12. A curve is defined by the polar equation

$$r = k\sin a\theta$$

where $a > 1$ and k is a positive constant.

a) State the maximum value of r in terms of k
b) State the minimum value of r in terms of k

State the number of times that the graph reaches the maximum distance from the pole on the curve giving your answer in terms of a for each of the following:

c) a is an odd integer for $0 \leq \theta \leq 2\pi$
d) a is an even integer for $0 \leq \theta \leq 2\pi$

Question 13. Sketch each of the following curves for the values of θ given, using a dashed line for $r < 0$:

a) $r = \cos 2\theta$
b) $r = \cos 3\theta$
c) $r = \cos 4\theta$
d) $r = \cos 5\theta$
e) $r = 3\cos 2\theta$
f) $r = 4\cos 3\theta$
g) $r = -\cos 4\theta$
h) $r = -3\cos 5\theta$

Question 14. A curve is defined by the polar equation

$$r = k\cos a\theta$$

where $a > 1$ and k is a positive constant.

a) State the maximum value of r in terms of k
b) State the minimum value of r in terms of k

State the number of times that the graph reaches the maximum distance from the pole on the curve giving your answer in terms of a for each of the following:

c) a is an odd integer for $0 \leq \theta \leq 2\pi$
d) a is an even integer for $0 \leq \theta \leq 2\pi$

Question 15. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, using a dashed line for $r < 0$:

a) $r = 2\sin 2\theta$

b) $r = 3\cos 2\theta$

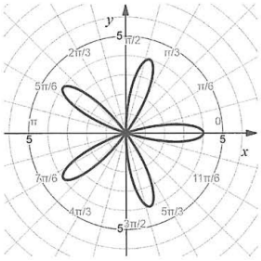
c) $r = 4\sin 3\theta$

d) $r = 2\cos 5\theta$

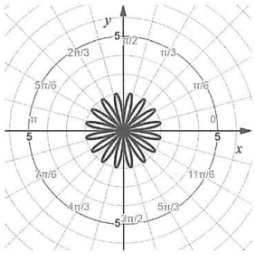
e) $r = \sin 6\theta$

Question 16. Find the polar equation in the form $r = k\cos a\theta$ or $r = k\sin a\theta$ where $k > 0$:

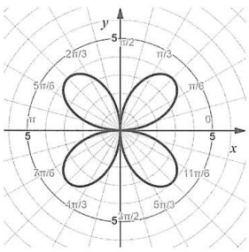
a)



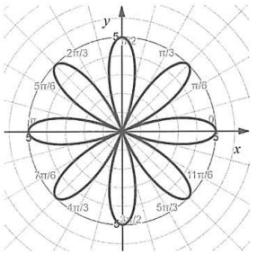
b)



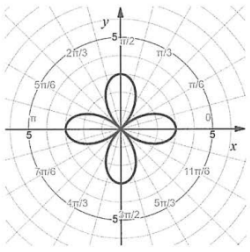
c)



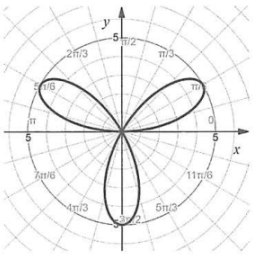
d)



e)



f)



Question 17. Sketch each of the following curves which are given in polar form:

a) $r = \frac{\theta}{2}$ for $0 \leq \theta \leq 2\pi$

b) $r = \frac{\theta}{2}$ for $-\pi \leq \theta \leq \pi$

Question 18. Given that $r = a\theta$ where a is a positive constant and for $0 \leq \theta \leq 2\pi$, describe what happens to the graph as a decreases.

Question 19. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = \frac{1}{\cos\theta}$ where $\theta \neq \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

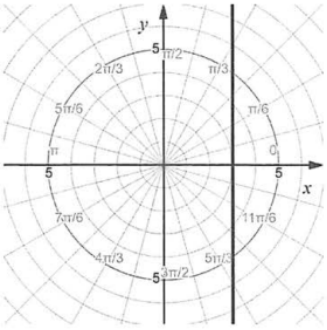
b) $r = \frac{1}{\sin\theta}$ where $\theta \neq \{0, \pi, 2\pi\}$

c) $r = \frac{5}{\cos\theta}$ where $\theta \neq \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

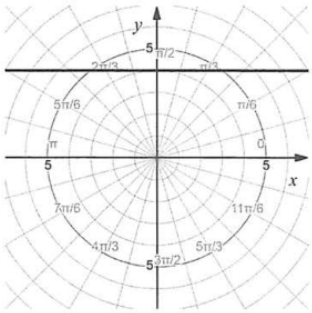
d) $r = -\frac{3}{\sin\theta}$ where $\theta \neq \{0, \pi, 2\pi\}$

Question 20. Find the polar equation of each of the following as r in terms of θ:

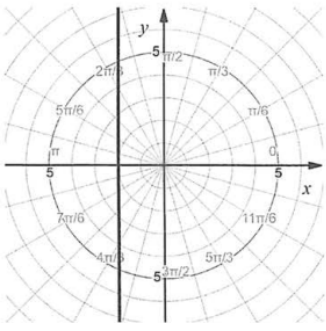
a)



b)



c)



Mix

Question 21. Sketch each of the following curves, using a dashed line for when $r < 0$:

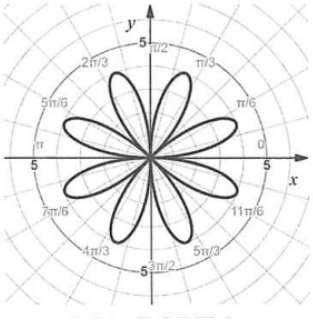
a) $r = 2\sin\theta$ for $0 \leq \theta \leq \pi$
b) $r = 2 + \cos\theta$ for $0 \leq \theta \leq 2\pi$
c) $r = 3\cos 4\theta$ for $-\pi \leq \theta \leq \pi$
d) $r = 2$ for $0 \leq \theta \leq 2\pi$
e) $r = 1 + 2\sin\theta$ for $0 \leq \theta \leq 2\pi$
f) $r = 3(1 + \cos\theta)$ for $0 \leq \theta \leq 2\pi$
g) $r = 2 - \sin\theta$ for $0 \leq \theta \leq 2\pi$
h) $r = \frac{3}{\cos\theta}$ for $\frac{\pi}{2} < \theta \leq \frac{3\pi}{2}$
i) $r = 5\sin 3\theta$ for $0 \leq \theta \leq \pi$
j) $r = 3 - 3\sin\theta$ for $0 \leq \theta \leq 2\pi$

Question 22. For constant $a > 0$, find the possible polar equation(s) for:

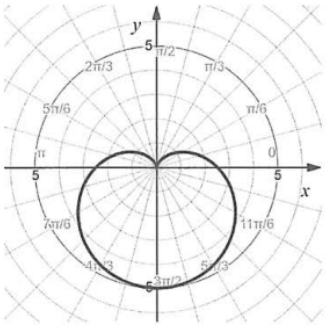
a) A line parallel to the x-axis at a distance a from the pole.
b) A line parallel to the y-axis at a distance $2a$ from the pole.
c) A circle with radius a with centre given by Cartesian coordinates $(\frac{a}{2}, 0)$
d) A circle with radius a with centre given by Cartesian coordinates $(0, a)$
e) A cardioid which passes through the pole, has a maximum value of r of $2a$ and passes through cartesian coordinates $(0, a)$
f) A cardioid which passes through the pole, has a maximum value of r of a and passes through cartesian coordinates $(\frac{a}{2}, 0)$

Question 23. Find the polar equation(s) of each of the following:

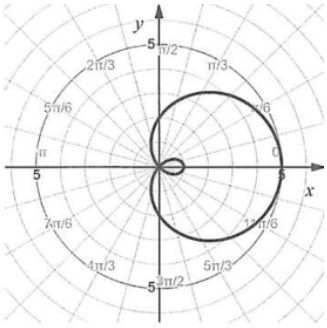
a)



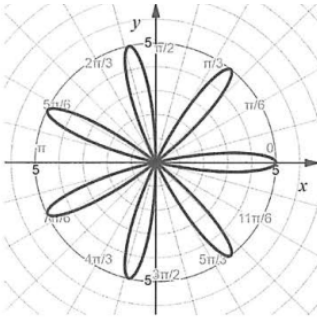
b)



c)



d)



Question 24. Sketch the graph of

$$r = \frac{2}{\sin(\theta - \frac{\pi}{6})}$$

for $\frac{\pi}{6} < \theta < \frac{7\pi}{6}$

Question 25. Sketch the graph of

$$r = \frac{3}{\cos(\theta - \frac{\pi}{3})}$$

for $-\frac{\pi}{6} < \theta < \frac{5\pi}{6}$ **Prerequisite knowledge for Q26: Hyperbolic Functions**

Question 26. For $0 \leq \theta \leq \pi$ sketch the polar graph of each of the following:

a) $r = \sinh\theta + \cosh\theta$

b) $r = \cosh\theta - \sinh\theta$

c) $r = \sinh\theta - \cosh\theta$

Prerequisite knowledge for Q27: Range and Domain of Functions

Question 27. Find the range of values of r for each of the following polar equations:

a) $r = 3\cos\theta$ for $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$
b) $r = 2\sin\theta$ for $0 < \theta \leq \frac{\pi}{2}$
c) $r = 3 + 2\cos\theta$ for $0 < \theta \leq \frac{\pi}{2}$
d) $r = \frac{3}{5+4\cos\theta}$ for $-\pi < \theta \leq \pi$
e) $r = \frac{3}{2-\sin\theta}$ for $0 < \theta \leq 2\pi$
f) $r = \frac{3}{1-2\cos\theta}$ for $0 \leq \theta \leq \frac{\pi}{2}$

Question 28. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, given that a is a positive constant. Identify the parts of the curve (if any) when $r < 0$ by using a dashed line. Label the point specified in terms of a:

a) $r = a$ and the point where $\theta = 0$
b) $r = a\cos\theta$ and the point where $\theta = 0$
c) $r = a\sin\theta$ and the point where $\theta = \frac{\pi}{2}$
d) $r = a\sin 2\theta$ and the point where $\theta = \frac{\pi}{4}$
e) $r = a\sin 3\theta$ and the point where $\theta = \frac{5\pi}{6}$
f) $r = a\cos 2\theta$ and the point where $\theta = 0$
g) $r = a\cos 3\theta$ and the point where $\theta = \frac{2\pi}{3}$
h) $r = a(1 + \sin\theta)$ and the point where $\theta = 0$
i) $r = a - a\cos\theta$ and the point where $\theta = \pi$

Question 29. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, given that a and b are positive constants and $a > b$. Identify when $r < 0$ by using a dashed line. Label the point where $\theta = 0$ in terms of a and b:

a) $r = a + b\sin\theta$
b) $r = b + a\sin\theta$
c) $r = a - b\sin\theta$
d) $r = a + b\cos\theta$
e) $r = b + a\cos\theta$
f) $r = b - a\cos\theta$

Intersection of Graphs

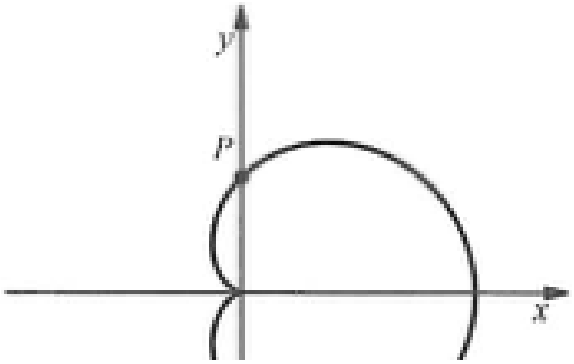
Question 30. For each of the following pairs of curves where $0 \leq \theta \leq 2\pi$, find the point of intersection giving your answer in the form (r, θ):

a) $r = 2\theta$ and $r = \theta + 2$
b) $r = 3\theta$ and $r = \theta + 1$
c) $r = \theta^2$ and $r = \theta + 2$

Question 31. The curve C has polar equation

$$r = 2(1 + \cos\theta)$$

for $-\pi < \theta \leq \pi$ is shown on the graph below:



a) Find the polar coordinates of the point P
b) State the polar coordinates of the point furthest away from the pole, Q
c) Find the points of intersection of curve C with a second curve given by

$$r = 3$$

for $-\pi < \theta \leq \pi$

Question 32. Find all the points of intersection of the curves

$$r = 2 - 3\cos\theta$$

and

$$r = 2\cos^2\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) where answers are given exact form.

Question 33. Find all the points of intersection of the curves

$$r = 2 - 3\sin\theta$$

and

$$r = 3\sin^2\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) to 3 significant figures. **Prerequisite knowledge for Q34: Double Angle Formulae**

Question 34. Find all the points of intersection of the curves

$$r = 4\sin 2\theta$$

and

$$r = 4\sin\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) where answers are given exact form.

Question 35.

a) Sketch the graph of

$$r = 6\cos 2\theta$$

for $0 \leq \theta \leq \pi$. Identify where the $r < 0$ by using a dashed line.

b) Find the points of intersection of the curve

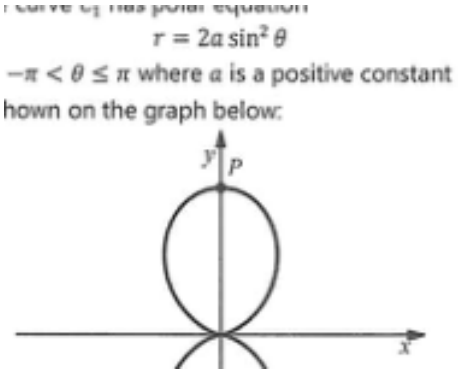
$$r = 6\cos 2\theta$$

for $0 \leq \theta \leq \pi$ and a circle of radius 3 centred at the pole.
Give your answers as polar coordinates (r, θ) where answers are given exact form and $r > 0$

Question 36. The curve C_1 has polar equation

$$r = 2a\sin^2\theta$$

for $-\pi < \theta \leq \pi$ where a is a positive constant is shown on the graph below:



a) Find the polar coordinates of the points P and Q giving your answer in terms of a

A second curve C_2 is given by

$$r = a(1 - \sin\theta)$$

for $-\pi < \theta \leq \pi$

b) Find the points of intersection of curve C_1 with C_2